Customer Shopping Behavior Analysis

End-to-end data analysis workflow uncovering patterns in purchasing, segmentation, and revenue trends

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Dataset Overview

3,900

Customer Records

Complete dataset with comprehensive shopping behavior data

18

Data Fields

Demographics, purchases, preferences, and behavior metrics

37

Missing Values

Review ratings imputed using median by category

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Data Pipeline

Data Cleaning

Excel to CSV, deduplication, missing value imputation

SQL Modeling

SQLite database, window functions, aggregations

Visualization

Power BI interactive dashboard with filters

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SQL Analysis Highlights

Revenue by Gender

- Male: $157,890
- Female: $75,191

Discount Impact

Customers using discounts spent above average, showing effective strategy

Product Ratings

- Gloves: 3.86
- Sandals: 3.84
- Boots: 3.82

Shipping Analysis

Express shipping: $60.48 average vs Standard: $58.46

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Customer Segmentation

Loyal Customers

3,476 customers with frequent purchases and high engagement

Returning Customers

424 customers with moderate purchase history

Subscription Impact

958 subscribed customers show higher repeat purchase rates

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Top Product Categories

1

Clothing

$104,264 revenue · Pants, Blouse, Shirt top performers

2

Accessories

Jewelry leads with 171 purchases · Sunglasses, Belt follow

3

Footwear

Sandals most popular · Shoes, Sneakers drive category sales

4

Outerwear

Jacket dominates · Coat, Sweater round out category

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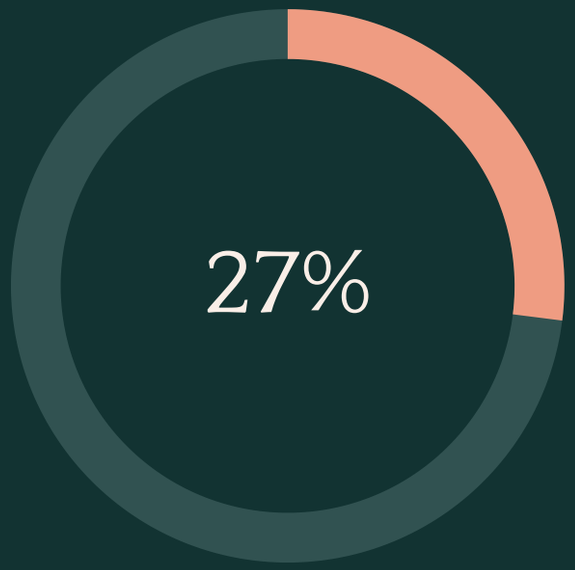
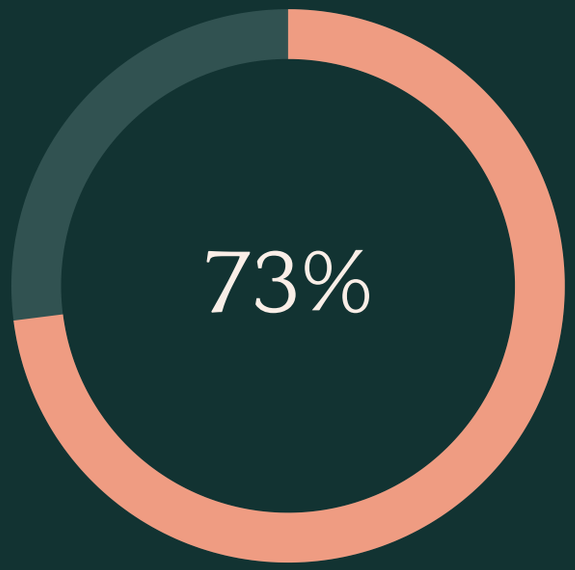
Demographic Insights



Subscribed

958 customers with subscription status



Non-Subscribed

2,942 customers without subscription

Revenue by Age

- Young Adults: $62,143
- Middle Aged: $59,197
- Adult: $55,978
- Senior: $55,763

Key Finding

Young adults drive highest revenue despite lower purchase frequency

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Power BI Dashboard

KPI Cards

Total revenue, average rating, customer count

Donut Charts

Gender, subscription, category distributions

Bar Charts

Revenue by category and age group analysis

Interactive Filters

Dynamic drill-down by gender, category, subscription

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Key Business Insights

High-Value Segments

Young adults and loyal customers drive revenue

Discount Strategy

Strategic discounts increase spending above average

Product Quality

High ratings indicate strong customer satisfaction

Subscription Value

Subscribers show higher engagement and repeat purchases

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Business Impact & Tools

Business Value

- Identify high-value customers
- Targeted marketing strategies
- Optimize pricing decisions
- Improve inventory focus
- Data-driven decision-making

Technology Stack

- Python (Pandas, NumPy)
- SQL (SQLite)
- Power BI
- Excel/CSV

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